

RESPONSE TO EDITOR

There are no whole truths in meta-analyses: all their truths are half-truths

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In a recent letter, Thomsen & Wernberg (2015) re-analyzed data compiled for our recent paper (Lyons *et al.*, 2014). In that paper, we examined the effects of macroalgal blooms and macroalgal mats on seven important measures of community structure and ecosystem functioning and explored several ecological and methodological factors that might explain some of the variation in the observed effects. Thomsen & Wernberg (2015) re-analyzed two small subsets of the data, focusing on experimental studies examining effects of blooms/mats on invertebrate abundance. Their analyses revealed two interesting patterns. First, they showed that macroalgal blooms reduced the abundance of communities that Thomsen and Wernberg categorized as ‘mainly infauna’, while increasing the abundance of communities categorized as ‘mainly epifauna’. Second, they showed that the impacts of macroalgal blooms on ‘mainly infauna’ communities increased with algal density in experiments that included multiple levels of algal density. These findings, as well as the conclusions that Thomsen & Wernberg (2015) draw from them, are largely consistent with our own expectations and interpretations. However, we also feel that some caution is required when interpreting the results of their analyses.

Ideally, syntheses comparing the effects of ecological phenomena on different subgroups would rely on a set of studies that had been designed for this specific aim. This is rarely possible, so synthetic works often rely on disparate studies that focus on the individual groups. None of the studies in the dataset analyzed by Thomsen & Wernberg (2015) overtly compared the effects of macroalgal blooms on infauna or epifauna, and only a proportion of them

explicitly focused on either of these groups individually. The sampling methods used by many studies in this dataset are likely to capture both infauna and epifauna living on the surface of the sediment, macroalgae, or other vegetation. Thus, classifying the communities investigated in these studies as ‘mainly infauna’ or ‘mainly epifauna’ requires subjective decisions that are difficult to make, and prone to error. We elected not to present a comparison of epifaunal and infaunal invertebrate communities in Lyons *et al.* (2014) because of these weaknesses, and the need to present a concise, coherent manuscript. Nevertheless, examining whether blooms affect infauna and epifauna differently is a useful and interesting exercise. Our dataset is likely the best currently available to make the comparison, and responding to Thomsen & Wernberg (2015) provides us with the opportunity to present our own broader analysis of the effects of macroalgal blooms on benthic marine invertebrates. Our analysis examined the effects of macroalgal blooms on both the abundance and species richness of benthic marine invertebrates, as measured by both experiments and observational studies (below).

Following the methods of Lyons *et al.* (2014), we conducted a mixed-model meta-analysis for invertebrate abundance and another for invertebrate species richness. We included study type (experimental vs. observational), invertebrate functional group (‘mainly’ infauna vs. epifauna, see Supporting information), and their interaction as explanatory variables in the analysis. This allowed us to directly assess the hypothesis that the observed effects of blooms and mats depend on each of these variables. For invertebrate abundance, there was very little support for the inclusion of the interaction term ($z = 0.518$, $P = 0.604$), so we present the results from a simplified model. Our analysis suggests that the impacts of blooms differ between epifauna and infauna (Fig. 1, $z = -2.596$, $P = 0.009$) and

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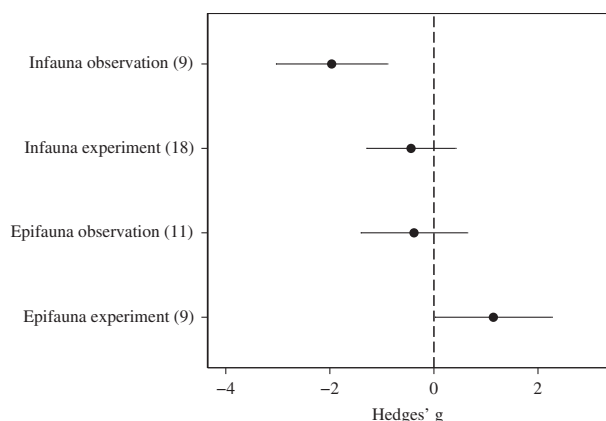


Fig. 1 Results of mixed-model meta-analysis estimating the effects of macroalgal blooms and mats on community measures of abundance (Hedge's g and 95% CI).

that observational studies tend to find effects that are more negative than experimental studies (Fig. 1, $z = -2.527$, $P = 0.012$). Like Thomsen & Wernberg (2015), we found a negative effect of blooms on infauna, and a positive effect of blooms on epifauna in experiments, but the magnitude of our summary effect size estimates was smaller, and the negative effect on infauna was nonsignificant. These disparities may be partially due to differences in the specific data included in each of the estimates. In several instances, Thomsen & Wernberg (2015) came to different conclusions about whether studies focused primarily on epifauna or infauna than we did. They also excluded some studies we chose to include in our analysis, and they used different data than we did when estimating the effect observed in some individual studies (see Supporting information). Differences in how meta-analyses were conducted may also contribute to the disparity between our results and those of Thomsen & Wernberg (2015). We used variance-weighted mixed models and used pooled estimates of the within-subgroup variance (recommended for subgroup analyses when sample sizes are relatively small, Borenstein *et al.*, 2009). Thomsen & Wernberg (2015) used separate, unweighted random-effects models for infauna and epifauna. They included multiple effect sizes from some of the multifactorial experiments in the study set. We always estimated a single effect size from multifactorial experiments. These differences affect the weight given to each study in the estimation of the effect sizes: Our approach gives less weight to studies with more uncertain effect sizes; their approach gives equal weights to all studies, except those for which they included multiple effect sizes.

For invertebrate richness, our analysis found a significant interaction between study type and invertebrate functional group affecting the observed effects

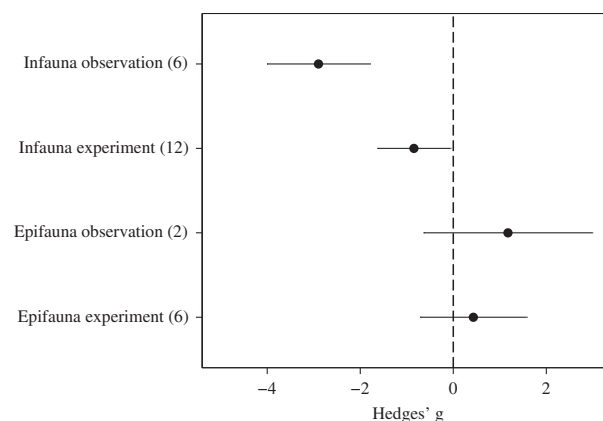


Fig. 2 Results of mixed-model meta-analysis estimating the effect of macroalgal blooms and mats on species richness (Hedge's g and 95% CI).

of macroalgal blooms ($z = -2.158$, $P = 0.031$). Both experimental and observational studies found nonsignificant positive effects of blooms on epifaunal species richness (Fig. 2). Both also found significant negative effects on infaunal species richness, but the effects observed in observational studies were more severe (Fig. 2).

Like Thomsen & Wernberg (2015), we found significant residual heterogeneity among studies (abundance: $Q = 1071.9$, $P < 0.001$; richness: $Q = 170.8$, $P < 0.001$), indicating that the effects of blooms are inconsistent, even after differences between subgroups (e.g., experiments examining epifauna, observational studies examining infauna) have been taken into account. There are many potential explanations for residual among-study heterogeneity, including variation in algal density, study duration, the size/extent of the bloom or mat, and methodological differences in how response variables were measured. It is also likely that both epifaunal and infaunal communities vary in their sensitivity to blooms. Members of both groups vary in their ability to tolerate anoxia and hydrogen sulfide (Riedel *et al.*, 2012). In addition, some infaunal species will move out of the sediment and into the overlying macroalgal mat when a bloom occurs (e.g., Österling & Pihl, 2001). This repositioning may help them to avoid hypoxic conditions while remaining within a bloom-affected area. It may also cause infaunal species to be collected and counted as epifauna in some studies. Conversely, some sessile epifaunal organisms are smothered by macroalgal accumulations, and epifauna living among the algae may die when rotting blooms induce hypoxia or release hydrogen sulfide. Thus, assemblages composed of different proportions of sensitive and resistant organisms will respond differently.

Thomsen & Wernberg (2015) consider differences between epifauna and infauna to be a 'devil in the detail' of Lyons *et al.*'s (2014) analysis comparing how communities of invertebrates, fish, bacteria, microalgae, macroalgae, seagrasses, or mixtures (of invertebrates and algae) respond to macroalgal blooms. However, pooling data from 'sensitive' and 'resistant' organisms within epifaunal and infaunal communities cancels out potentially important contrasts in a similar way that pooling studies of epifaunal and infaunal invertebrate communities does. We point this out because the existence of this 'devil in the detail' of analyses intended to 'promote more nuanced conclusions' about the effects of blooms illustrates the tension inherent in ecological synthesis and raises important questions. With ecological processes and outcomes contingent on so many factors, which differences do we pay attention to? In the context of ecological meta-analyses, which effects can be meaningfully combined? The answers should be determined by the ecological question or management problem of interest, but are likely to be influenced by the availability of sufficient data, as well as the opinions and interests of the researcher. For instance, we chose not to calculate an overall effect size synthesizing all of the community-level responses in our study because we question the meaningfulness of an effect size that includes studies of disparate responses such as species richness, organism abundance, and benthic community respiration. Others have estimated such effect sizes, and they might argue they are both meaningful and useful.

The density, size, and duration of macroalgal blooms and mats are likely to play a very important role in determining the nature and magnitude of their effects. Nearly two decades ago, Raffaelli *et al.* (1998) suggested that much of the variation in blooms' effects is due to differences in the intensity and size of macroalgal blooms and that rigorous definition of these factors would improve our understanding by facilitating comparisons between blooms. However, many studies of blooms' effects lack detailed information about algal densities and the spatial extent of blooms, and when information is available, it is recorded in such a way that comparisons are difficult (e.g., measures of % cover, wet mass per area, dry mass per area, thickness of the algal mat for measures of algal density) (Lyons *et al.*, 2014). The paucity of data prevented us from conducting a broad analysis of how these factors alter the impacts of blooms on marine ecosystems, and is reflected in the fact that Thomsen & Wernberg (2015) are forced to rely on just four studies in their second analysis. Their results provide an indication that the positive effect of macroalgal blooms on epifaunal

abundance is larger at higher algal densities. However, it is unclear whether the increasing effect of blooms with increased algal density is linear, nonlinear, or part of a more complex, nonmonotonic relationship. This question, along with the broader one of how algal density alters blooms' other effects, will be answered more easily if researchers record algal density information (preferably in wet or dry mass per unit area) or directly study this specific question more often.

Despite our minor cautions, concerns, and preference for our own methodology, we largely agree with the decisions, findings, and interpretations of Thomsen & Wernberg (2015), and we are glad they took the initiative to look more deeply into the data we compiled. We feel that their analyses, and the additional analyses we present above, are interesting and potentially useful to ecosystem managers. However, these analyses do not alter our interpretation of blooms' effects on invertebrate communities: If we combine all of the available evidence, it appears that blooms have a negative effect, but that effect is highly variable (Lyons *et al.*, 2014). Nor do we see variation due to different invertebrate functional groups or sensitivities, algal abundance, or other untested factors as 'devils in the details'. This term has pejorative connotations: referring to a troublesome part of a larger whole, or a detail with potential to mar something larger if not handled correctly. We have always seen such factors as opportunities for future work. And we discussed many of them in Lyons *et al.* (2014). The goal of meta-analysis is to synthesize and summarize in order to describe general patterns, and meta-analysts must 'average over' many of the details that make studies different from one another to accomplish this goal. Rather than worry about devils in details when we use meta-analysis, it is more important that we treat the generalizations they provide appropriately, remembering that 'There are no whole truths: all truths are half-truths. It is trying to treat them as whole truths that plays the devil' (Whitehead, 1954).

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Data S1. Supplemental methods, data, and references..

Table S1. Effect sizes (Hedge's g) and variances used in the analysis of macroalgal blooms and mats effects on invertebrate abundance.

Table S2. Effect sizes (Hedge's g) and variances used in the analysis of macroalgal blooms and mats effects on invertebrate species richness.